



UNIT 07

ANALYTICAL CHEMISTRY



ANALYTICAL CHEMISTRY

It is the branch of science in which we qualify and quantify the matter **OR** which deals with the qualitative and quantitative analysis (تحليلية) of matter.

TYPES OF ANALYSIS

QUALITATIVE ANALYSIS

It is the identification of an element in a compound.

Further Divisions of Qualitative Analysis: Qualitative analysis can be further divided into two main approaches: inorganic qualitative analysis and organic qualitative analysis.

1. INORGANIC QUALITATIVE ANALYSIS:

Deals with the identification of presence of different inorganic elements, compounds, ions etc.



LINK TO LEARNING

<https://youtu.be/kkBFG1mTSBk?si=luXfTpgykWhk1wQ8>

LINK 01



Using litmus paper

Wet blue litmus paper (see page 134) can be used to test for the presence of chlorine gas.

1. Hold wet blue litmus paper above a test tube that holds chlorine gas.

2. The litmus paper turns red at first, indicating that chlorine is acidic.

3. Chlorine gas then bleaches the red litmus paper white.

Test tube containing chlorine gas.

2. ORGANIC QUALITATIVE ANALYSIS:

Deals with the identification of presence of different organic elements, compounds, ions etc.



Molisch's test detects carbohydrates. A few drops of Molisch's reagent (α -naphthol in alcohol) are added to the sample, followed by concentrated sulfuric acid. A purple or violet ring at the interface indicates the presence of carbohydrates, formed by a reaction between furfural and α -naphthol.

QUANTITATIVE ANALYSIS

It is the quantify of one or more substance present in a compound or sample. They are classifying as physical Quantitative Analysis or chemical Quantitative Analysis.

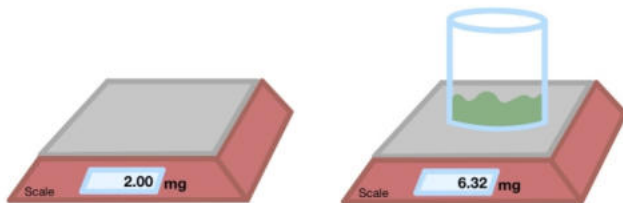
ERROR

Error is the difference between the true value of a quantity and the measured value of that quantity. In analytical chemistry, errors can be classified into two types: systematic errors and random errors.

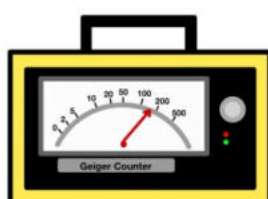
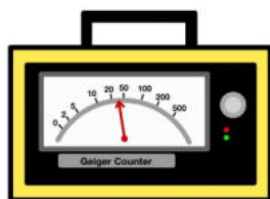
1. SYSTEMATIC ERROR OR DETERMINATE ERROR

This is caused by the improper analytical method or by malfunctioning in the instrument.

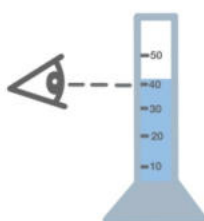
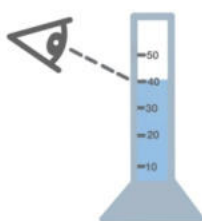
EXAMPLE: If a balance is always reading 10 grams too high, this is a systematic error that can be corrected by subtracting 10 grams from all of the measurements made with that balance.

**Instrumental Error**

The scale is improperly calibrated so it reads 2mg even with nothing on it. This causes the the beaker to appear 2mg heavier than it would be, and any other measurements would read 2mg heavier than what they actually weigh

**Environmental Error**

The Geiger counter shows the radiation, however, when a cell phone is near the Geiger counter, the radiation is read as greater than what it actually is because the cell phone is radiating RF waves which disrupts the reading

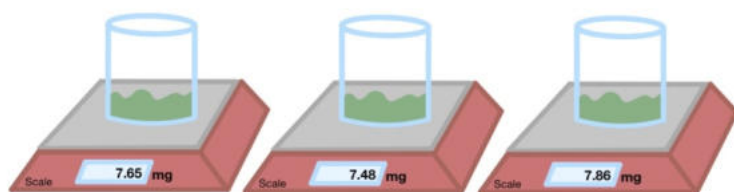
**Observational Error**

The observer looks at the beaker from above eye level and write down the volume as 40mL, when in reality when looked at eye level you can see the liquid has a volume of about 42mL

2. RANDOM ERROR

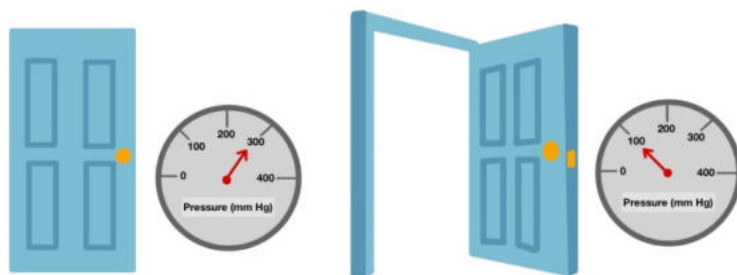
Random errors, also known as indeterminate errors, are occur due to variable methods or due to the limitations of instruments.

EXAMPLE: If a balance is not perfectly calibrated, this will introduce random errors into the measurements.



Observational Error

A student repeatedly weights a beaker with solution and gets different measurements every time



Environmental Error

When getting a pressure reading, students were entering and exiting the lab, causing the pressure to be lower

ACCURACY

Accuracy refers to how close a measured value is to the true or accepted value. It quantifies the degree of correctness or the absence of systematic measurement errors. Accuracy is determined by comparing the measured value to a reference value or a known standard.

PRECISION

Precision refers to the degree of consistency, reproducibility, or exactness of multiple measurements of the same quantity. It quantifies the level of random errors or variations in measurement.

CLASSICAL METHODS

Classical methods typically involve qualitative and quantitative analysis based on visual observations and chemical reactions. They often utilize basic laboratory glassware, such as beakers, flasks, pipettes, and burettes, along with fundamental laboratory instruments like balances and pH meters.

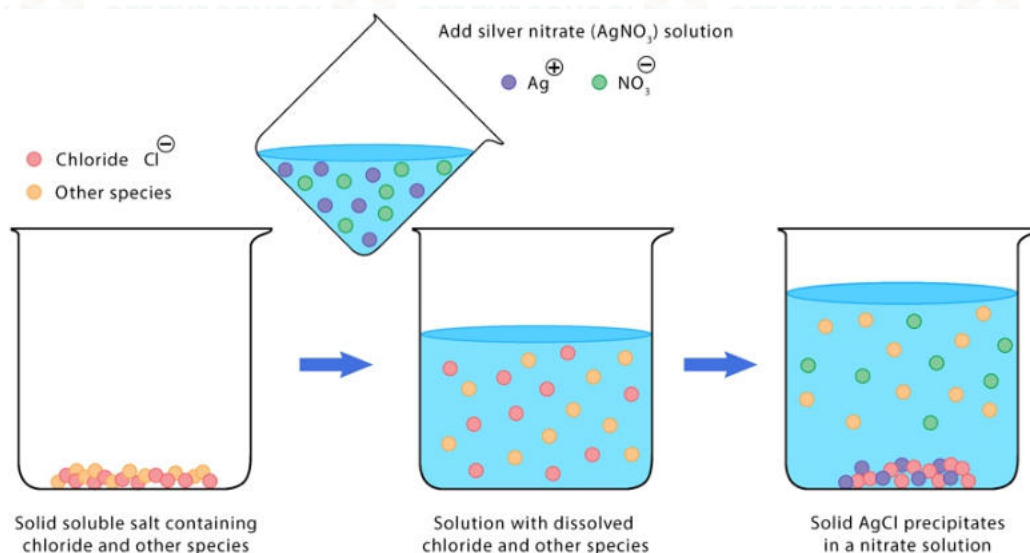
Examples of classical methods include gravimetric analysis, titration, precipitation reactions, colorimetry, and distillation.

WHAT IS GRAVIMETRIC ANALYSIS?

Gravimetric analysis is a method in analytical chemistry to determine the quantity of an analyte based on the mass of a solid.

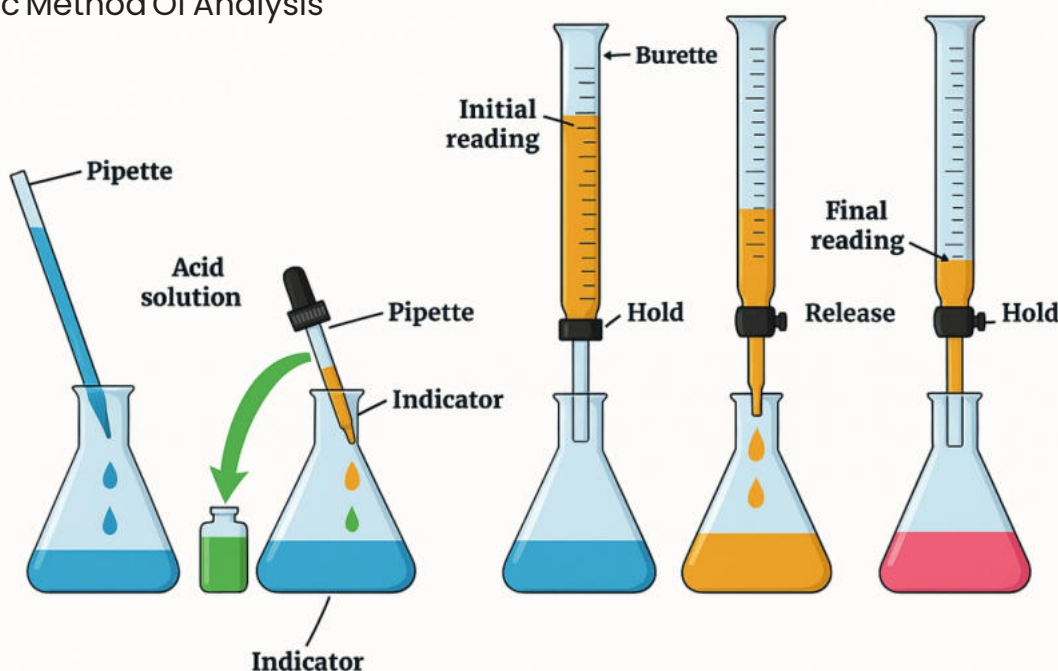
STEPS FOLLOWED IN THE GRAVIMETRIC ANALYSIS

1. Preparation of a solution containing a known weight of the sample.
2. Separation of the desired constituent.
3. Weighing the isolated constituent.
4. Computation of the amount of the particular constituent in the sample from the observed weight of the isolated substance.



TITRIMETRIC ANALYSIS

A Volumetric Method Of Analysis



Titration is a classical analytical technique used to determine the concentration of an analyte in a solution by reacting it with a solution of known concentration (titrant). The reaction between the analyte and the titrant is monitored until it reaches the equivalence point, where stoichiometrically equivalent amounts of the analyte and titrant have reacted.

EXAMPLE:

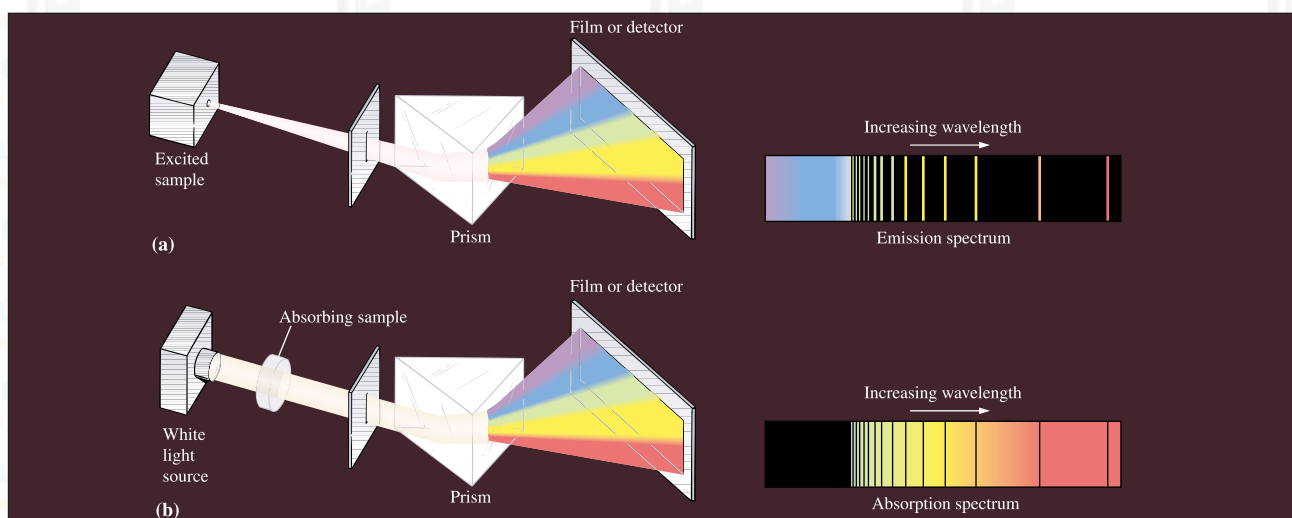
An example of titration analysis is the determination of the concentration of acid (e.g., hydrochloric acid, HCl) in a given solution using a standardized solution of sodium hydroxide (NaOH) as the titrant.

The steps involved in this titration are as follows:

1. Measure and place the acid solution in a flask.
2. Add a few drops of an indicator (e.g., phenolphthalein or methyl orange).
3. Gradually add standardized NaOH from a burette while swirling.
4. Stop when the indicator shows a color change (endpoint).
5. Record the volume of NaOH used.
6. Use the balanced equation and titration data to calculate the acid concentration.

SPECTROSCOPY

Interaction of matter with a radiative energy as function of wavelength or frequency.



Some common spectroscopic methods include:

1. UV-VISIBLE SPECTROSCOPY

First of all we understand the term Ultraviolet (UV) radiation: is a type of electromagnetic radiation with a wavelength shorter than visible light but longer than X-rays. Visible radiation on the other hand, refers to the portion of the electromagnetic spectrum that can be seen by the human eye, with wavelengths ranging from approximately 400 to 700 nanometers.

1. ULTRAVIOLET SPECTROSCOPY

Ultraviolet-visible spectroscopy or ultraviolet-visible spectrophotometry (UV-Vis or UV/Vis) refers to absorption spectroscopy or reflectance spectroscopy in the ultraviolet-visible spectral region. Ultraviolet-visible (UV-VIS) spectroscopy is an analytical method that can measure the analyte quantity depending on the amount of light received by the analyte.

2. INFRARED SPECTROSCOPY

Infrared (IR) spectroscopy measures the absorption or transmission of infrared light, which corresponds to molecular vibrations and rotations. It provides information about the functional groups and chemical bonds present in a compound, aiding in the identification and characterization of organic and inorganic molecules.

CHROMATOGRAPHY

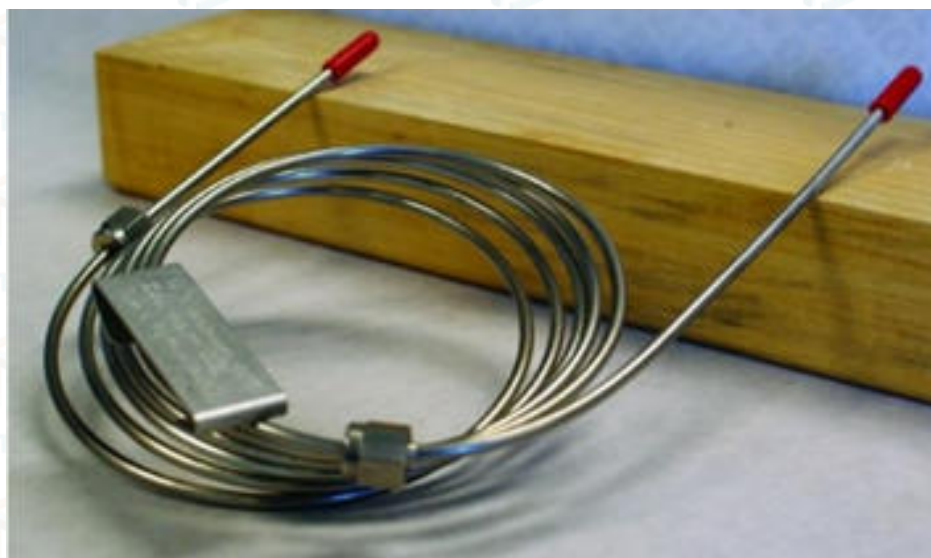
(Greek = chroma "color" and graphein "writing"): Chromatography is a laboratory technique used to separate and analyze different substances in a mixture.

GAS CHROMATOGRAPHY (GC)

Gas Chromatography (GC) is a laboratory technique used to separate and analyze volatile compounds in a mixture. Here's an overview of the instruments involved and how they work:

1. INJECTOR

The sample is first vaporized and then injected into the GC instrument using an injector.



2. COLUMN

The sample then moves through a long, narrow column filled with a stationary phase, which separates the components of the mixture based on their boiling points or vapor pressures.

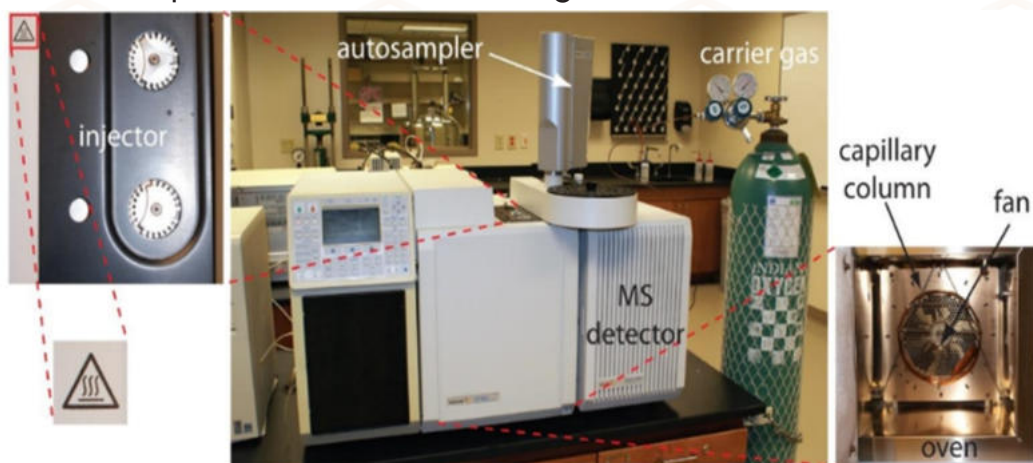
3. MOBILE PHASE

The mobile phase, usually an inert gas such as helium or nitrogen, carries the vaporized sample through the column.

4. DETECTOR

After separation, the individual components are detected by a detector, which measures the quantity of each component. Common detectors used in GC include:

► **FLAME IONIZATION DETECTOR (FID):** measures the ionization of the components by a hydrogen flame and produces an electrical signal.



► **ELECTRON CAPTURE DETECTOR (ECD):** measures the decrease in electron activity of the components as they pass through a radioactive ionization source and produce an electrical signal.

► **MASS SPECTROMETER (MS):** ionizes the components and measures their masses to determine their identity.

5. DATA SYSTEM

The signals from the detector are then processed by a data system, which converts them into a chromatogram, a graph showing the detector response over time.

ELECTROCHEMICAL METHODS

Electrochemical methods in analytical chemistry use electrical potentials to measure the concentration of species in a sample.

Examples include:

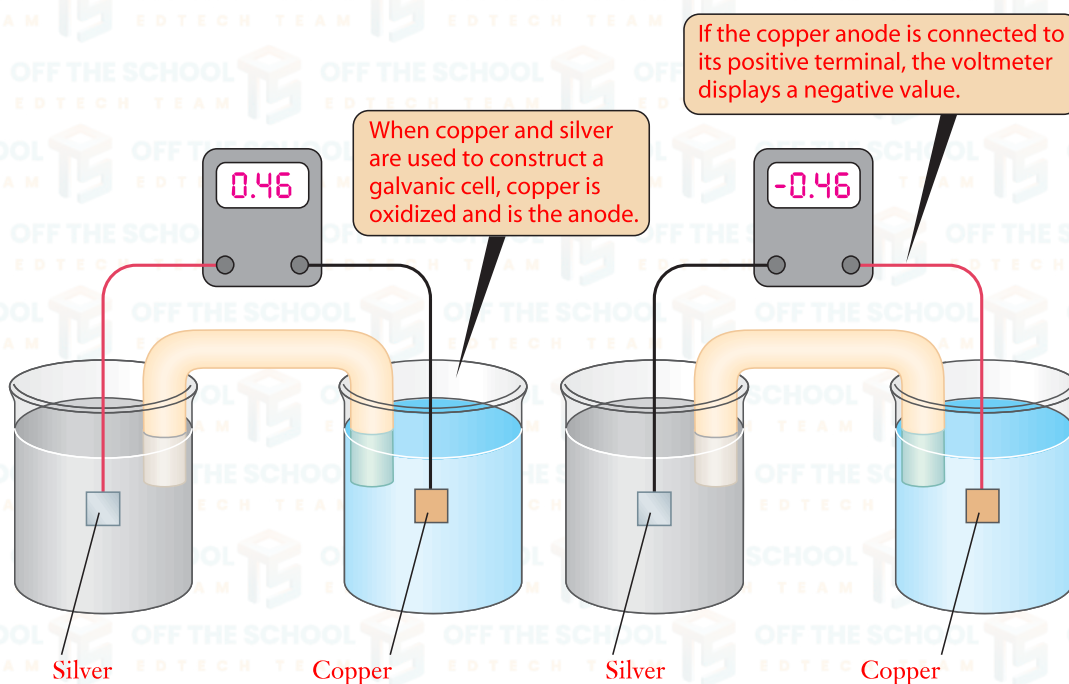
- ✓ Voltammetry
- ✓ Amperometry
- ✓ Potentiometry
- ✓ Conductometry
- ✓ Coulometry

These techniques are often used for analysis of biological, environmental, and industrial samples. An electrochemical cell is a device that uses electrical energy to produce a chemical reaction or generates electrical energy from a chemical reaction.

- ▶ It has two electrodes, an anode and a cathode, placed in an electrolyte solution.
- ▶ The electrodes are connected to an external circuit that allows the flow of electrons.
- ▶ The flow of electrons generates a potential difference between the electrodes, causing ions in the electrolyte to move and produce a chemical reaction.
- ▶ Salt Bridge provides the passage for ions movement without mixing the solutions.
- ▶ In a battery, the cell generates electrical energy through the reaction between the electrodes.

POTENTIOMETRY

Potentiometry is a type of electrochemical analysis that measures the potential difference (voltage) between two electrodes in a solution to determine the concentration of a particular species in the sample. It is based on the principle that the potential difference is directly proportional to the concentration of the species in the solution.



The positive terminal of the meter is connected with a red wire and the negative with a black wire in this figure.

CONDUCTOMETRY

Conductometry is a type of analytical technique in electrochemistry that measures the electrical conductivity of a solution to determine the concentration of ions in the sample.

Characteristics	Classical Methods	Instrumental Methods
Basis	Reliance on chemical reactions and separations.	Reliance on electronic instruments and devices.
Type of Analysis	Often qualitative analysis based on visual observations.	Capable of both quantitative and qualitative analysis.
Equipment	Traditional glassware (e.g., beakers, flasks, burettes).	Advanced instruments (e.g., spectrophotometers, chromatographs, mass spectrometers).
Automation	Limited automation; manual operations.	Higher automation with computer-controlled systems.
Examples	Gravimetric analysis, titration, colorimetry, precipitation reactions.	Spectroscopy (UV-Visible, IR, NMR), chromatography (HPLC, GC), mass spectrometry, electrochemical analysis.

We cannot solve our problems with the same thinking we used when we created them.

Albert Einstein

